

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T56 · Probability Toolkit

With Answers

Questions with full worked solutions

■ 37 pages

FAMILY MAP

Probability Toolkit

F01 Calculate theoretical single-event probability

11 Q | 34 marks

F02 Use expected frequency

6 Q | 19 marks

Working Unit 18

4MA1-2H Q11

ORIGINAL QUESTION**3 marks**

11 Khalid has a box of counters.

17 of the counters are red

28 of the counters are blue

the rest of the counters are orange

Khalid is going to take at random a counter from the box.

The probability that Khalid will take an orange counter is $\frac{4}{9}$

Work out the number of orange counters that are in the box.

(Total for Question 11 is 3 marks)

Worked Solution 18

Recover a population from a probability

UNIT 1
3 marks**Non-orange counters**

$$17 + 28 = 45, \quad P(\text{non-orange}) = 1 - \frac{4}{9} = \frac{5}{9}.$$

Total and orange

$$N = 45 \div \frac{5}{9} = 81, \quad 81 - 45 = 36.$$

Final answer

36 orange counters

Question 01

4MA1-1H Q2

ORIGINAL QUESTION

4 marks

- 2 450 students were asked how they travelled to school on Monday.
Each student walked or travelled by bus or travelled by car or travelled by bicycle.
Each student used just one method of travel.

One of these students is chosen at random.

The table shows information about the probability of each method of travel.

Method of travel	walk	bus	car	bicycle
Probability	0.20	x	$2x$	0.26

Work out how many of the 450 students travelled by car.

Worked Solution 01

Probability to frequency

MODULAR UNIT 1

4 marks

Complete the probability total

$$0.20 + x + 2x + 0.26 = 1 \Rightarrow x = 0.18.$$

Car frequency

$$P(\text{car}) = 0.36, \quad 450(0.36) = 162.$$

Final answer

162 students

Question 06

4MA1-1H Q4 (a)

ORIGINAL QUESTION UNIT 1

2 marks

- 4 A biased spinner has three sections each of a different colour.

The table shows the probability that, when the spinner is spun once, it will land on blue or on orange or on white.

Colour	blue	orange	white
Probability	0.58	$2x$	x

- (a) Work out the value of x

$x = \dots\dots\dots$
(2)

Worked solution 06

Complete the spinner probabilities

MODULAR UNIT 1

2 marks

(a) – Complete the spinner probabilities

Use total probability

$$0.58 + 2x + x = 1 \Rightarrow 3x = 0.42.$$

Solve

$$x = 0.14.$$

Final answer

$$x = 0.14$$

Question 07

4MA1-1H Q4 (b)

ORIGINAL QUESTION UNIT 1

2 marks

- 4 A biased spinner has three sections each of a different colour.

The table shows the probability that, when the spinner is spun once, it will land on blue or on orange or on white.

Colour	blue	orange	white
Probability	0.58	$2x$	x

The spinner is spun 250 times.

- (b) Work out an estimate for the number of times the spinner will land on blue.

.....
(2)

Worked solution 07

Expected blue outcomes

MODULAR UNIT 1

2 marks

(b) – Expected blue outcomes

Scale the blue probability

$$250(0.58) = 145.$$

Final answer

145

Question 13

4MA1-2H Q2

ORIGINAL QUESTION

3 marks

- 2 A biased spinner can land on green or on yellow or on brown or on pink.

The table gives the probabilities that, when the spinner is spun, it will land on green or on yellow or on brown.

Colour	green	yellow	brown	pink
Probability	0.32	0.13	0.28	

Timucin spins the spinner 200 times.

Work out an estimate for the number of times the spinner lands on pink.

Worked Solution 13

Expected frequency from a spinner

MODULAR UNIT 1

3 marks

Find pink probability

$$P(\text{pink}) = 1 - (0.32 + 0.13 + 0.28) = 0.27.$$

Scale to 200 spins

$$0.27 \times 200 = 54.$$

Final answer

54

Original question 21

4MA1-2H Q5 M01

MODULAR UNIT 1

4 marks

- 5 Cody has a biased 5-sided spinner, numbered 1, 2, 3, 4, 5

The table gives the probabilities that when the spinner is spun it will land on 2 or on 3 or on 5

Number	1	2	3	4	5
Probability		0.14	0.17		0.21

The probability that the spinner will land on 1 is the same as the probability that the spinner will land on 4

Cody is going to spin the spinner 400 times.

Work out an estimate for the number of times the spinner will land on 4

.....

Worked solution 21

Estimate a spinner frequency

MODULAR UNIT 1

4 marks

M01 – Estimate a spinner frequency**Use total probability**

$$0.14 + 0.17 + 0.21 + x + x = 1 \Rightarrow 2x = 0.48.$$

Find the missing probability

$$x = 0.24.$$

Scale to 400 spins

$$400(0.24) = 96.$$

Final answer

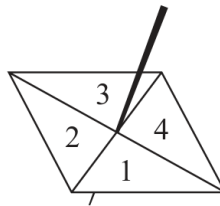
96

Original question 01

4MA1-1H Q2 Q2

ORIGINAL QUESTION**4 marks**

- 2 Here is a biased 4-sided spinner.



The table gives the probabilities that, when the spinner is spun once, it will land on 1 or it will land on 3

Number	1	2	3	4
Probability	0.26		0.18	

The probability that the spinner will land on 2 is equal to the probability that the spinner will land on 4

Ravina is going to spin the spinner a number of times.

Ravina works out that an estimate for the number of times the spinner will land on 3 is 45

Work out an estimate for the number of times the spinner will land on 4

Worked solution 01

Chapter: Probability Toolkit

MODULAR UNIT 1

4 marks

Q2 – Chapter: Probability Toolkit

Family: Use expected frequency • Variant: Reverse expected frequency to find trials or probability

MethodLet the probability of landing on 2, and also on 4, be p .*Why: The question states that these two probabilities are equal.***Method** $2p + 0.26 + 0.18 = 1$, so $2p = 0.56$ and $p = 0.28$.*Why: All four spinner probabilities add to 1.***Method**If 45 is the estimated frequency for outcome 3, the estimated number of spins is $45/0.18 = 250$.*Why: Estimated frequency equals probability multiplied by the number of trials.***Method**The estimated frequency for outcome 4 is $250 \times 0.28 = 70$.*Why: Apply the probability of outcome 4 to the same number of spins.***Final answer****estimated frequency: 70**

Original question 23

4MA1-2H Q6

ORIGINAL QUESTION**5 marks**

- 6 A bag contains only pink sweets, white sweets, green sweets and red sweets.

The table gives each of the probabilities that, when a sweet is taken at random from the bag, the sweet will be green or the sweet will be red.

Sweet	pink	white	green	red
Probability			0.2	0.35

The ratio

number of pink sweets : number of white sweets = 2 : 1

There are 28 red sweets in the bag.

Work out the number of white sweets in the bag.

Worked solution 23

Chapter: Probability Toolkit

MODULAR UNIT 1

5 marks

Q6 – Chapter: Probability Toolkit

Family: Calculate theoretical single-event probability • Variant: Use category ratios in a probability model

MethodThere are 28 red sweets with probability 0.35, so total sweets $= 28 / 0.35 = 80$.*Why: Frequency equals probability times total.***Method**Pink plus white probability $= 1 - 0.2 - 0.35 = 0.45$.*Why: All four probabilities sum to 1.***Method**With pink:white=2:1, white probability is $0.45 / 3 = 0.15$.*Why: Split the 0.45 in the ratio 2:1.***Method**White sweets $= 80 \times 0.15 = 12$.*Why: Multiply the probability by the total.***Method**

Therefore the bag contains 12 white sweets.

*Why: State the requested count.***Final answer****white sweets: 12**

- 4 The table shows information about the weights, in kilograms, of 40 babies.

Weight (w kg)	Frequency
$2 < w \leq 3$	12
$3 < w \leq 4$	16
$4 < w \leq 5$	9
$5 < w \leq 6$	2
$6 < w \leq 7$	1

- (a) Write down the modal class.

One of the 40 babies is going to be chosen at random.

- (c) Find the probability that this baby has a weight of more than 5 kg.

(2)

Worked solution

January 2020 | Question 4 | 2 marks

Family: Calculate theoretical single-event probability • Variant: Use favourable outcomes over total outcomes

Method / working

1. Weights above 5 kg occur with frequency $2+1=3$.
2. Probability= $3/40$ (not $3/10$).

Final answer: $3/40$

- 6 A bag contains only red beads, blue beads, green beads and yellow beads.

The table gives the probabilities that, when a bead is taken at random from the bag, the bead will be blue or the bead will be yellow.

Colour	red	blue	green	yellow
Probability		0.24		0.31

The probability that the bead will be green is twice the probability that the bead will be red.

Sofia takes at random a bead from the bag.

She writes down the colour of the bead and puts the bead back into the bag.

She does this 180 times.

Work out an estimate for the number of times she takes a red bead from the bag.

Worked solution

January 2020 | Question 6 | 4 marks

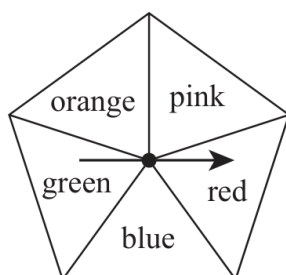
Family: Use expected frequency • Variant: Calculate expected frequency from probability and trials

Method / working

1. Let red probability be r ; green is $2r$.
2. $r+0.24+2r+0.31=1$, so $r=0.15$.
3. Expected red count= 180×0.15 .
4. Estimate= 27 red beads.

Final answer: 27 red beads

6 Grace has a biased 5-sided spinner.



Grace is going to spin the arrow on the spinner once.

The table below gives the probabilities that the spinner will land on red or on blue or on green.

Colour	Red	Blue	Green	Orange	Pink
Probability	0.20	0.12	0.08		

The probability that the spinner will land on orange is 3 times the probability that the spinner will land on pink.

(a) Work out the probability that the spinner will land on orange.

(3)

Worked solution

June 2021 | Question 6 | 3 marks

Family: Calculate theoretical single-event probability • Variant: Complete a biased probability table using total one

Method / working

1. The remaining probability is $1 - (0.20 + 0.12 + 0.08) = 0.60$.
 2. Orange: pink = 3:1, so orange is three quarters of 0.60.
 3. Orange probability = 0.45.
- Final answer: 0.45

Grace spins the arrow on the spinner 150 times.

(b) Work out an estimate for the number of times the spinner lands on blue.

(2)

Worked solution

June 2021 | Question 6 | 2 marks

Family: Use expected frequency • Variant: Calculate expected frequency from probability and trials

Method / working

1. Use expected frequency = probability x number of spins.
2. $0.12 \times 150 = 18$ blue results.

Final answer: 18

- 3 There are 54 fish in a tank.
Some of the fish are white and the rest of the fish are red.

Jeevan takes at random a fish from the tank.

The probability that he takes a white fish is $\frac{4}{9}$

- (a) Work out the number of white fish originally in the tank.

.....
(2)

Jeevan puts the fish he took out, back into the tank.

He puts some more white fish into the tank.

Jeevan takes at random a fish from the tank.

The probability that he takes a white fish is now $\frac{1}{2}$

- (b) Work out the number of white fish Jeevan put into the tank.

.....
(2)

Worked solution

November 2021 | Question 3 | 2 marks

Family: Calculate theoretical single-event probability | Variant: Recover population composition from a single-event probability

METHOD / WORKING

1. (a) Let the number of white fish be W .

$$2. \frac{W}{54} = \frac{49}{9}$$

$$3. W = 54 \times \frac{49}{9} = 294$$

FINAL ANSWER

(a) 24 white fish

- 3 There are 54 fish in a tank.
Some of the fish are white and the rest of the fish are red.

Jeevan takes at random a fish from the tank.

The probability that he takes a white fish is $\frac{4}{9}$

- (a) Work out the number of white fish originally in the tank.

.....
(2)

Jeevan puts the fish he took out, back into the tank.

He puts some more white fish into the tank.

Jeevan takes at random a fish from the tank.

The probability that he takes a white fish is now $\frac{1}{2}$

- (b) Work out the number of white fish Jeevan put into the tank.

.....
(2)

Worked solution

November 2021 | Question 3 | 2 marks

Family: Calculate theoretical single-event probability | Variant: Recover population composition from a single-event probability

METHOD / WORKING

1. (b) There are still 54 fish, and now
2. $\frac{\text{white fish}}{54} = \frac{1}{2}$
3. white fish = 27
4. Jeevan added
5. $27 - 24 = 3$
6. Answer: (a) 24 white fish, (b) 3 white fish.

FINAL ANSWER

(b) 3 white fish.

- 2 Toy cars are made in a factory.
The toy cars are made for 15 hours each day.
5 toy cars are made every 12 seconds.

For the toy cars made each day, the probability of a toy car being faulty is 0.002

Work out an estimate of the number of faulty toy cars that are made each day.

.....
(Total for Question 2 is 4 marks)

Answer reference | May-Nov 2020 | Question 2

Family: Calculate theoretical single-event probability • Variant: Recover population composition from a single-event probability

2	$15 \times 60 \times 60 (= 54\,000)$ oe or $\frac{60}{12} \times 60 \times 15 (= 4500)$ oe or $5 \times \frac{60}{12} \times 60 (= 1500)$ oe		4	M1	M2 for $\frac{15 \times 60 \times 60 \times 5}{12}$ (= 22 500)
	'54000' $\div 12 \times 5 (= 22\,500)$ oe or '4500' $\times 5 (= 22\,500)$ oe or '1500' $\times 15 (= 22\,500)$ oe			M1	
	'22 500' $\times 0.002$ oe			M1	dep on M2 for a complete method
		45		A1	
					Total 4 marks

Worked solution

May-Nov 2020 | Question 2 | 4 marks

Family: Calculate theoretical single-event probability • Variant: Recover population composition from a single-event probability

Solution

15 hours = $15 \times 60 \times 60 = 54000$ seconds

The machine makes 5 cars every 12 seconds, so it makes

$(54000)/(12) \times 5 = 22500$

cars.

Expected faulty cars:

$22500 \times 0.002 = 45$

Answer: 45.

- 7 Some members of a library were asked to name the type of book that they each liked to read the best.

One of the members is chosen at random.

The table shows information about the probability of the type of book that this member answered.

Type of book	comedy	romance	mystery	thriller
Probability	0.24	0.40	$3x$	x

48 members answered comedy books.

Work out how many of the members answered mystery books.

(Total for Question 7 is 4 marks)

Answer reference | January 2023 | Question 7

Family: Calculate theoretical single-event probability • Variant: Complete a biased probability table using total one

7	$1 - (0.24 + 0.4) (= 0.36)$ oe or $3x + x = 1 - (0.24 + 0.4)$ oe		4	M1
	$48 \div 0.24 (= 200)$ or "0.36" $\div 4 (= 0.09)$ or "0.36" $\div 4 \times 3 (= 0.27)$			M1
	"0.27" $\times$ "200" or "200" $\times$ "0.36" $\div 4 \times 3$ ("200" $- 48 -$ "80") $\div 4 \times 3$			M1 for a complete method
		54		A1
				Total 4 marks

7 ALT	$1 - (0.24 + 0.4) (= 0.36)$ oe or $3x + x = 1 - (0.24 + 0.4)$ oe		4	M1
	$48 \div 24 (= 2)$ oe or $\left(\frac{"0.36"}{4} \times 3\right) \div 0.24 \left(= \frac{9}{8} = 1.125\right)$ oe or $\left(\frac{"36"}{4} \times 3\right) \div 24 \left(= \frac{9}{8} = 1.125\right)$ oe			M1
	"2" $\times$ $\left(\frac{"36"}{4} \times 3\right)$ oe or " $\frac{9}{8}$ " $\times 48$ oe or ("27" $\div 24$) $\times 48$ oe			M1 for a complete method
	Correct answer scores full marks (unless from obvious incorrect working)	54		A1
				Total 4 marks

Worked solution

January 2023 | Question 7 | 4 marks

Family: Calculate theoretical single-event probability • Variant: Complete a biased probability table using total one

Official final mark-scheme pages are embedded immediately after this question.

- 9 Mark has three bags, bag A, bag B and bag C

Bag A contains 120 counters.

The probability of taking at random a red counter from bag A is 0.45

Bag B contains 80 counters.

The probability of taking at random a red counter from bag B is 0.3

Bag C is empty.

Mark puts all the counters from bag A and bag B into bag C

Mark takes at random a counter from bag C

Work out the probability that he takes a red counter.

(Total for Question 9 is 3 marks)

Worked solution

November 2025 | Question 9 | 3 marks

Family: Calculate theoretical single-event probability • Variant: Recover population composition from a single-event probability

Solution

Bag A has

$$120 \times 0.45 = 54$$

red counters.

Bag B has

$$80 \times 0.30 = 24$$

red counters.

After combining the bags, the total number of red counters is

$$54 + 24 = 78$$

The total number of counters is

$$120 + 80 = 200$$

$$P(\text{red}) = (78)/(200) = 0.39$$

Answer: 0.39.

PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION